

Predictive Power vs. Interpretability

A Bayesian Network Approach to Churn Prediction — Overview

CUSTOMERS

7,043

Telco Churn dataset

CHURN RATE

26.5%

1,869 churned

BEST MODEL (ROC-AUC)

0.845

Logistic Regression

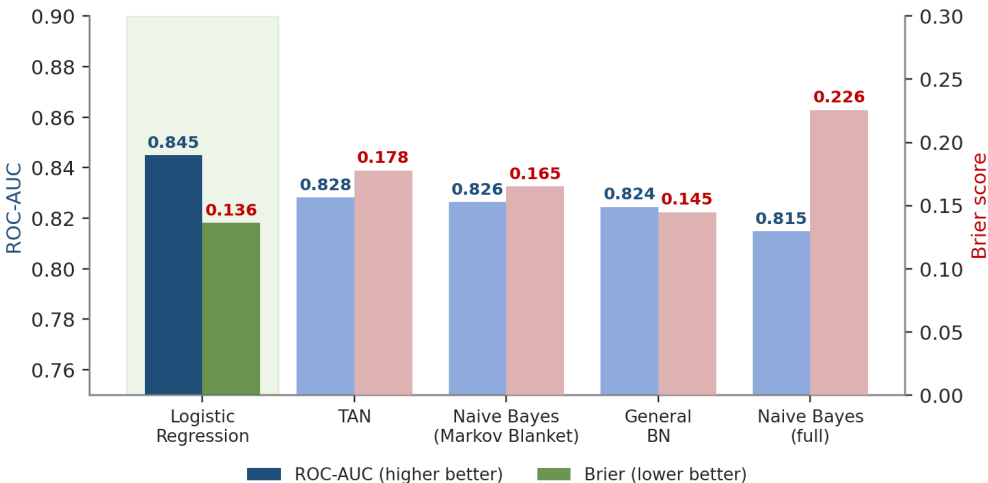
MODELS COMPARED

5

LogReg, NB, TAN, General BN

Research question: Does model complexity (Logistic Regression → Naive Bayes → TAN → General Bayesian Network) actually improve predictive performance — and does any gain justify the loss of interpretability?

Model Comparison — Test Set (2,113 customers)



Headline Finding

Logistic Regression won on every metric — highest ROC-AUC (0.845), highest PR-AUC, and best calibration (Brier 0.136). None of the three Bayesian Network variants closed the gap, despite up to 38 structural edges and a dedicated structure-learning step.

This was confirmed statistically, not just numerically: paired bootstrap testing (1,000 resamples) shows the ROC-AUC gap is real in all three comparisons — see page 2.

Data Quality Check

11 of 7,043 records (0.16%) had a blank TotalCharges value stored as whitespace, invisible to standard null checks. Root cause: all 11 were brand-new customers (tenure=0), not yet billed. Fixed via domain logic (TotalCharges=0), not median imputation.

Methodology: Escalating Model Complexity

1. Logistic Regression

Linear baseline. 30 one-hot features. No graph structure.

2. Naive Bayes

Assumes all features independent given Churn. Fast, fully interpretable.

3. TAN

Tree-Augmented Naive Bayes — each feature gets one extra parent besides Churn.

4. General BN

Freely learned structure (HillClimbSearch) — up to 3 parents per node.

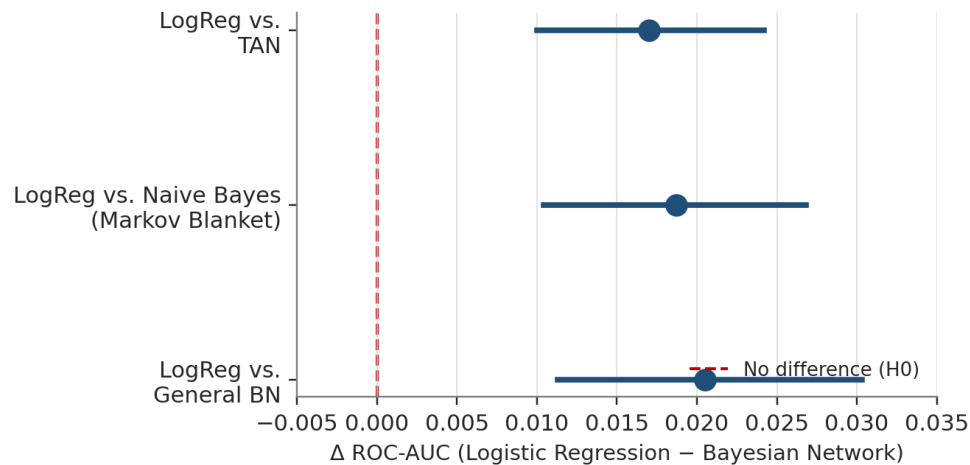
Predictive Power vs. Interpretability

Is the Difference Real? Statistical Testing & Structural Complexity

Predictive Power vs. Interpretability

Page 2 of 3

All 3 Comparisons: 95% CI Excludes Zero



Pre-Registered Hypothesis Test

H0: no significant ROC-AUC difference between Logistic Regression and each Bayesian Network variant. Tested via paired bootstrap (1,000 resamples on the identical 2,113-customer test set) — chosen over 5x2cv to avoid re-learning BN structure 10 times per comparison.

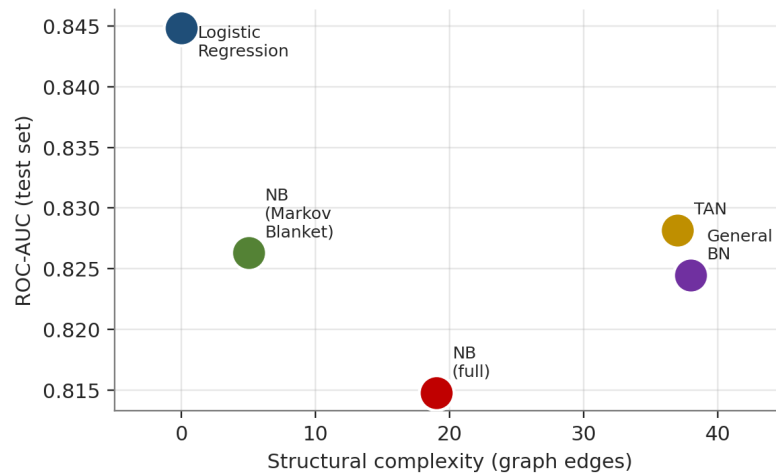
VS. TAN
+0.017
95% CI: [0.010, 0.024]

VS. NB (MARKOV BLANKET)
+0.019
95% CI: [0.010, 0.027]

VS. GENERAL BN
+0.021
95% CI: [0.011, 0.030]

All 3 comparisons: H0 rejected — the gap is statistically real, in every case

More Edges Did Not Mean Better Discrimination



Complexity Table

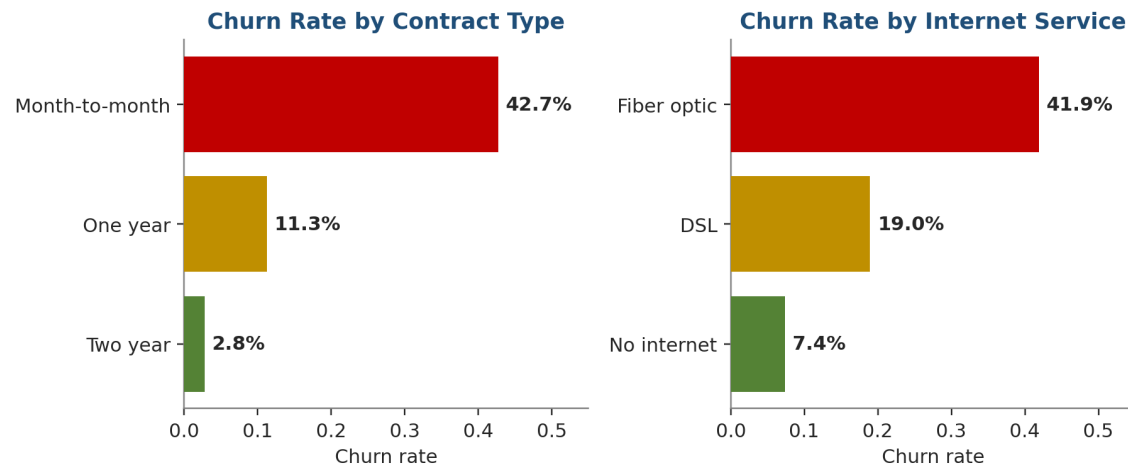
Model	Edges	Max Parents	Markov Blanket
Logistic Regression	—	—	—
Naive Bayes (full)	19	1	19
Naive Bayes (MB)	5	1	5
TAN	37	2	19
General BN	38	3	5

Predictive Power vs. Interpretability

Churn Patterns, the Markov Blanket & Business Recommendation

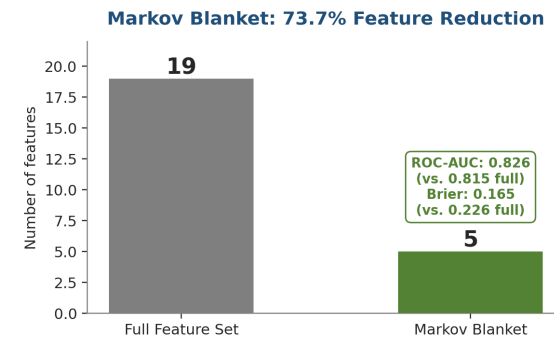
Predictive Power vs. Interpretability

Page 3 of 3



Markov Blanket = Independently Confirmed Variables

General BN's freely-learned structure reduced Churn's Markov Blanket to just 5 variables — matching exploratory analysis with no prior guidance: Contract, InternetService, PaperlessBilling, TechSupport, tenure. 'gender' was excluded entirely from the structure.



Business Recommendation

Deploy Logistic Regression: no Bayesian Network variant significantly beat it, so the added complexity (up to 38 edges) isn't justified. But keep Markov Blanket analysis as a lightweight, model-agnostic feature-selection diagnostic — it independently re-derived the variables most strongly linked to churn, with 73.7% fewer features.

Limitations

- HillClimbSearch finds a local optimum — a different seed or scoring method could yield a different structure.
- Numeric features were quantile-binned for Bayesian models but left continuous for Logistic Regression — a natural, not fully neutral, comparison.
- Network edges reflect statistical dependency, not confirmed causation.

What Makes This Analysis Rigorous

- ✓ H0 pre-registered before testing, not after seeing results
- ✓ Same 2,113-customer test set scored once, for all 5 models
- ✓ Natural class distribution preserved (no artificial balancing) so probabilities stay genuine
- ✓ Calibration checked, not just discrimination (ROC-AUC alone would have missed the Brier gap)

Actionable Retention Levers

- Month-to-month contracts churn at 15x the rate of two-year contracts (42.7% vs. 2.8%) — the single highest-leverage retention target.
- Fiber optic customers churn at 41.9% vs. 7.4% with no internet service — worth investigating price, competition, or service-quality factors.
- Churn risk concentrates in the first months of tenure — an onboarding-period retention push is likely to have outsized impact.